

# Largest Triangle

| Limits         |         | Submit                                                 |  |
|----------------|---------|--------------------------------------------------------|--|
| ⌚ Time         | 1000 ms | Code<br><div></div> <div>C++ ▾</div> <div>Submit</div> |  |
| 💾 Memory       | 512 MiB |                                                        |  |
| <> Source size | 64 KiB  |                                                        |  |
| ☑ Points       | 100     |                                                        |  |

| Verdict               | Time | Memory | Language | Points | Samples |
|-----------------------|------|--------|----------|--------|---------|
| No submissions yet :( |      |        |          |        |         |

## Description

You are given  $N$  points in 2D coordinate system, all of which have integer coordinates.

Among all of the triangles whose vertices are among the given points, you must find the largest one by area.

Area of a triangle whose vertices are points  $P_1(x_1, y_1)$ ,  $P_2(x_2, y_2)$ ,  $P_3(x_3, y_3)$  is calculated using the formula:

$$2 \cdot \text{Area}(P_1 P_2 P_3) = |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Notice that the area multiplied by 2 is an integer, so you will be asked to **print the largest area multiplied by 2** so that the output is an integer.

Below is a short code sample how the formula would look in C++.

```
struct Point {
    long long x;
    long long y;
};

long long DoubleArea(Point A, Point B, Point C){
    return abs(A.x * (B.y - C.y) + B.x * (C.y - A.y) + C.x * (A.y - B.y));
}
```

Be careful to use **long long** to avoid overflow.

## Input Data

In the first line, there is an integer  $3 \leq N \leq 100000$ , the number of points.

In each of the next  $N$  lines there are two integers  $0 \leq x_i, y_i \leq 200000$ , each pair representing the coordinates of the  $i$ -th point.

## Output Data

In the first and only line, print one integer: the largest area out of all the triangles formed by the given  $N$  points **multiplied by 2**.

## Scoring

First test data cluster is worth 10 points and  $3 \leq N \leq 250$ .

Second test data cluster is worth 10 points and  $3 \leq N \leq 600$ .

Third test data cluster is worth 30 points and the points are generated randomly (uniformly).

Fourth test data cluster is worth 30 points and  $3 \leq N \leq 6600$ .

Fifth test data cluster is worth 20 points and  $3 \leq N \leq 12000$ .

## Example

**Input:**

```
3
0 0
200000 200000
200000 0
```

**Output:**

```
400000000000
```

### Explanation:

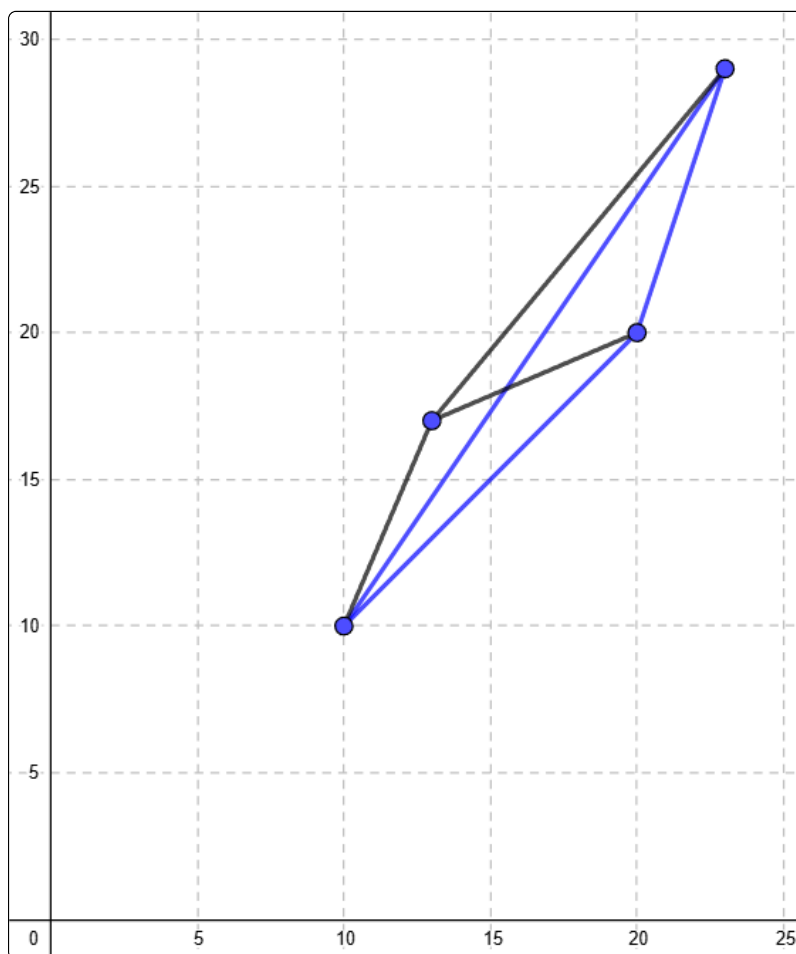
There is only one triangle of area 200000000000. Its double area is 400000000000 and that is the answer.

### Input:

```
4
10 10
20 20
13 17
23 29
```

### Output:

```
60
```



### Explanation:

The blue triangle formed by points (10, 10), (20, 20) and (23, 29) has the largest area, which is 30.0, so the answer is 60.

